
4. Troubleshooting

4-1. Checkpoints by Error Mode

Oscilloscope Setting Values	
Voltage/DIV	1V/div
TIME/DIV	500ms/div

4-1-1. No Power

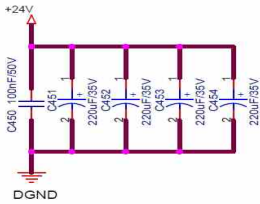
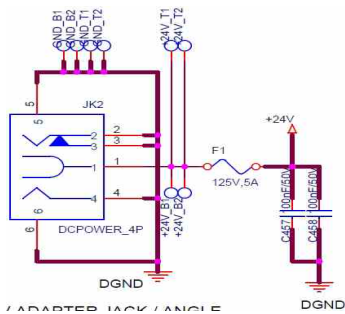
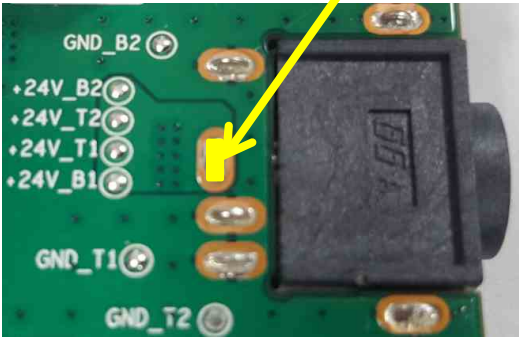
Symptom	- If you connect the power cables, the LED on the front will turn on - If the units do not work
Major checkpoints	- If the cable is improperly connected or the motherboard does not work, the LED on the front is not working. In this case, check the following. - Check the cable connection. - Check the fuse of the components.
Diagnostics	<pre> graph TD Start([The power does not turn on.]) -- Yes --> J1{Check if 24V is measured on the MAIN PCB JK2 pin #1?} J1 -- Yes --> J2{Check if "H(3.3V)" is measured at the pin 248 BGA of the U104 when MAIN PCB power is turned on.} J1 -- No --> J3{Check if 24V is measured at the Adapter pin #1.} J2 -- Yes --> A1[Change the MAIN PCB.] J2 -- No --> A2[Replace The U104] J3 -- Yes --> A3[MAIN PCB JK2 change.] J3 -- No --> A4[Replace OLED Module] A1 --> J4{Check the AC Cord output.} A2 --> J4 A3 --> J4 A4 --> J4 J4 -- No --> A5[Replace the AC Cord.] J4 -- Yes --> A6[Check Adapter and replace Adapter.] </pre>
Caution	Before working, disconnect the power cord.

■ Location of Parts

UIC104_248 BGA
ALIVE_3.3V
R832/R833



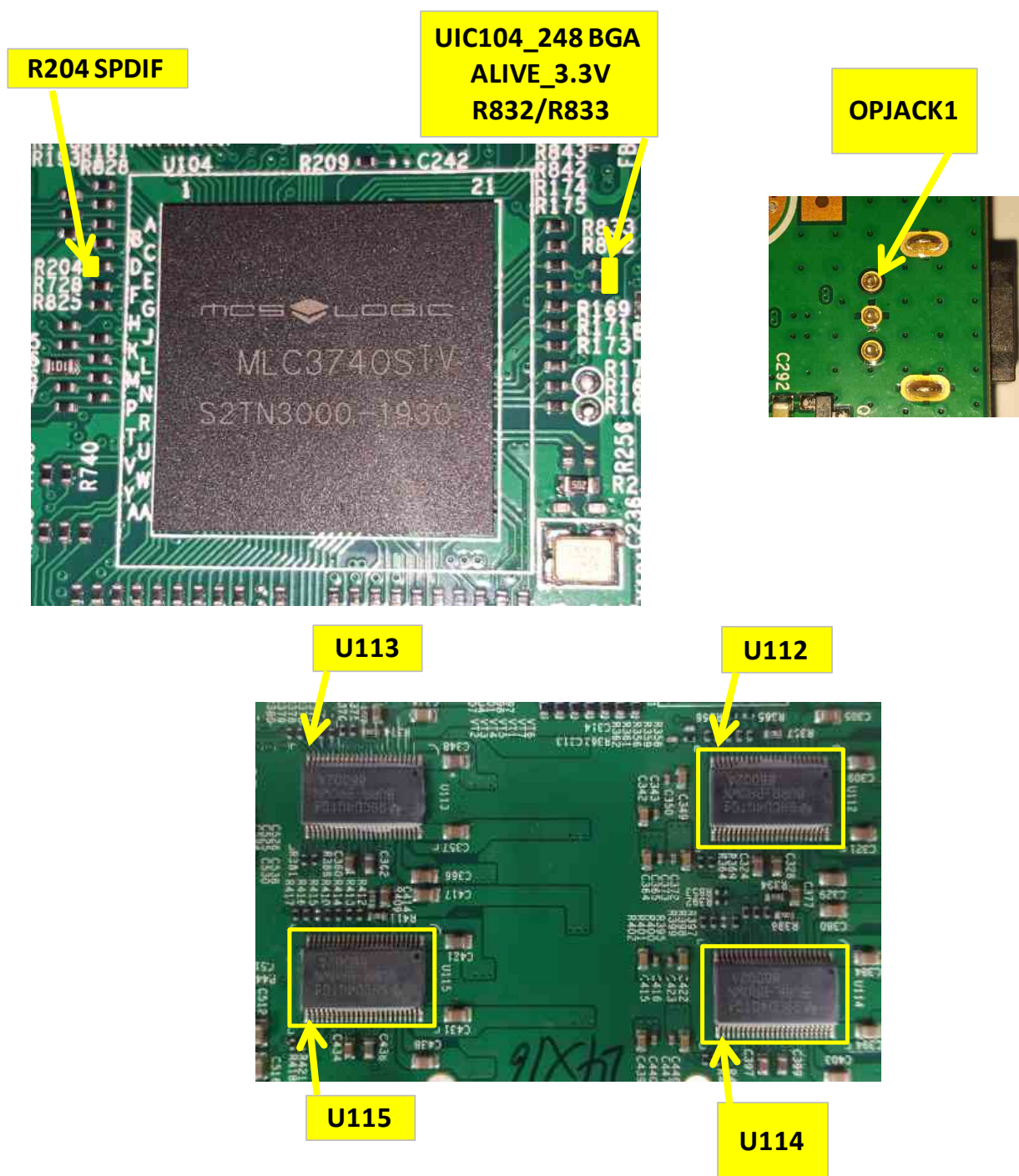
JK2
DC_24V



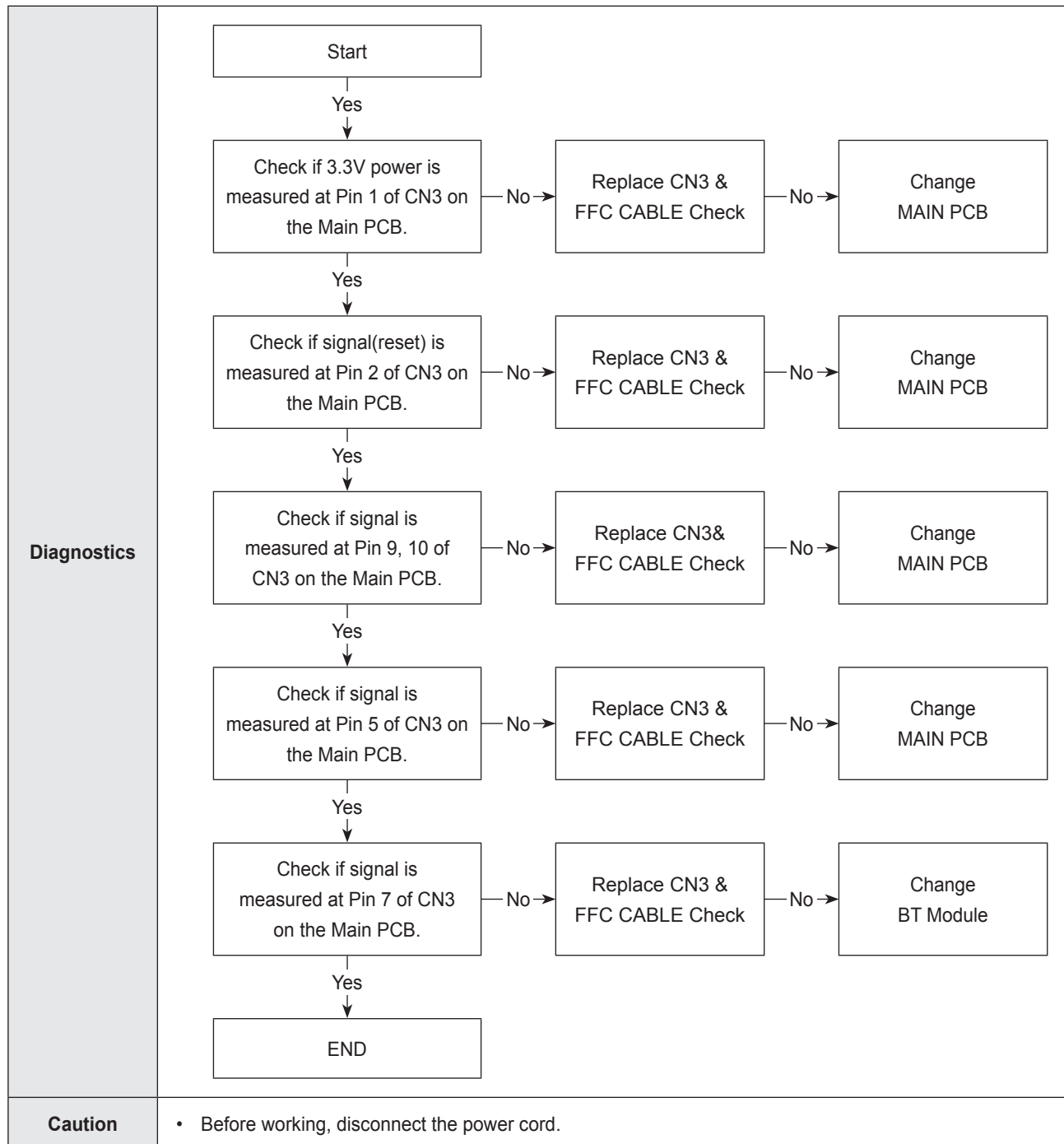
4-1-2. No Sound Output

Symptom	If the sound is not output.
Major checkpoints	- Speaker connector is not connected properly or check for damage. - Make sure your sound processing functions operate properly on the motherboard. - Make sure the speaker isn't bad.
Diagnostics	<pre> graph TD Start([No Sound Output]) --> CheckSPDIF[Check SPDIF Output.] CheckSPDIF --> Q1{Check if 3.3V power is measured at UIC104_248 BGA ALIVE_3.3V R832/R833, R204 SPDIF on the MAIN PCB?} Q1 -- No --> A1[Change the MAIN PCB.] Q1 -- Yes --> Q2{Check if 3.3V power is measured at pin 1 of OPJACK1?} Q2 -- No --> A2[Change the MAIN PCB.] Q2 -- Yes --> CheckDSP[DSP Output Check.] CheckDSP --> Q3{Check if signal is measured at RP106_7pin of U112, U113, U114, U115 on the MAIN PCB?} Q3 -- No --> A3[Replace IC.] A3 -- No --> A4[Change the MAIN PCB.] Q3 -- Yes --> Q4{Check if signal is measured at pin RP102_7, 5pin, R184, R183, R198, R740, R208 of U112, U113, U114, U115 on the MAIN PCB?} Q4 -- No --> A5[Replace IC] A5 -- No --> A6[Change the MAIN PCB.] Q4 -- Yes --> CheckSpeaker[Speaker output check.] CheckSpeaker --> Q5{Check the MAIN PCB ACN1, ACN2 speaker output signal?} Q5 -- No --> A7[Change speaker connector.] A7 -- No --> A8[Change the MAIN PCB.] Q5 -- Yes --> A9[Change Speaker L/R.] </pre>
Caution	Before working, disconnect the power cord.

■ Location of Parts

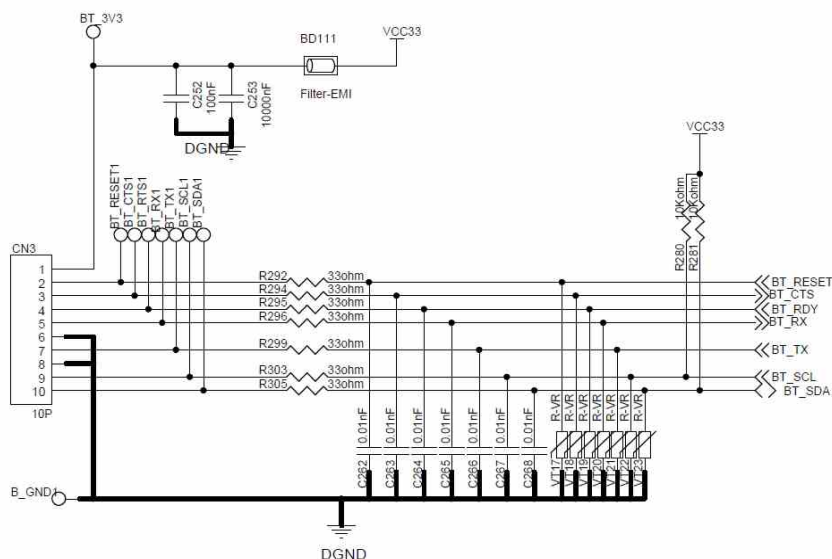


4-1-3. BT Module Failure Analysis



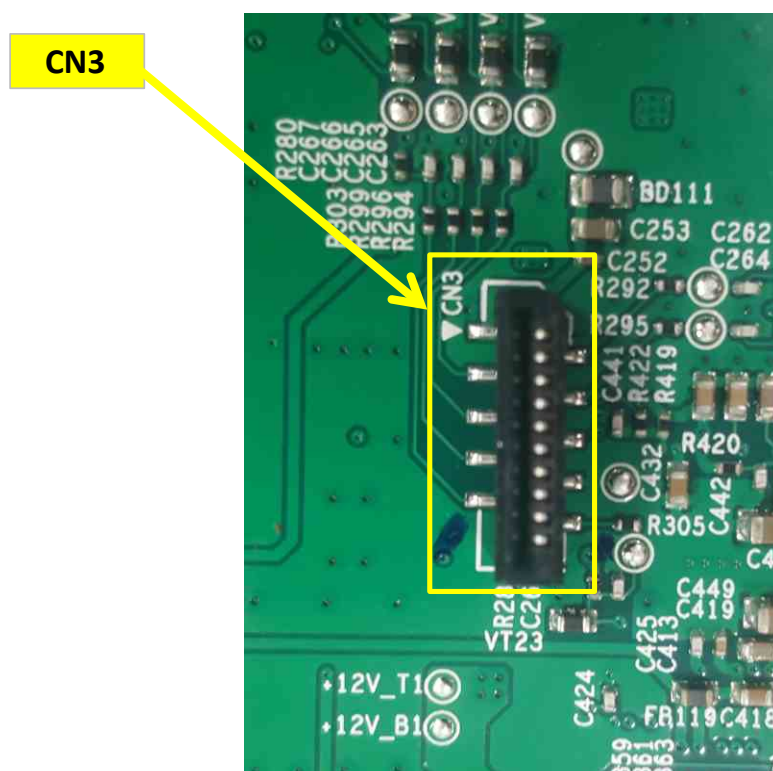
■ Location of Parts

Bluetooth 4.2 Module

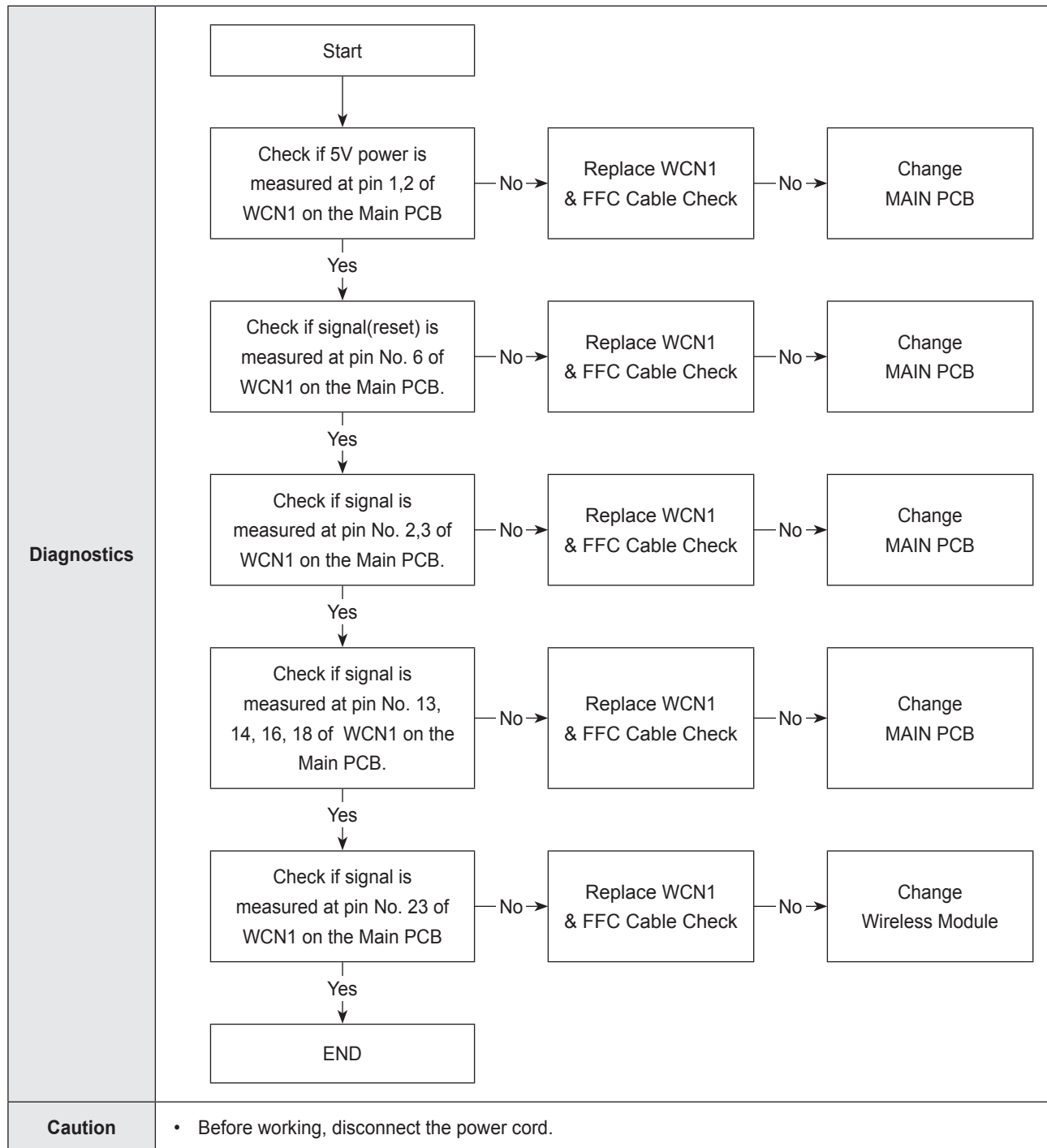


TO MB8811Q-A B/T MODULE

CN3	
MAIN <-> BT	
1	BT_3V3
2	BT_RESET
3	BT_CTS
4	BT_RTS
5	BT_RX
6	DGND
7	BT_TX
8	DGND
9	BT_SCL
10	BT_SDA

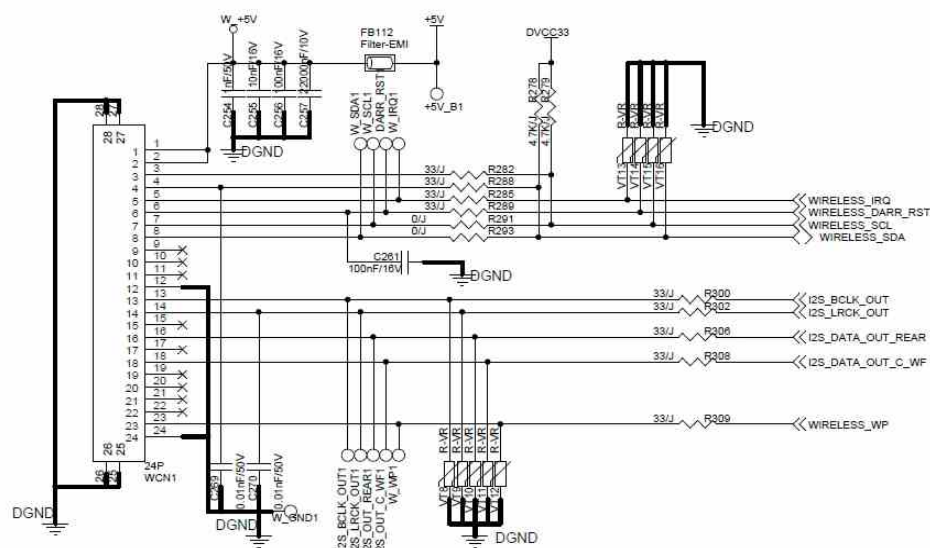


4-1-4. Wireless Module Failure Analysis

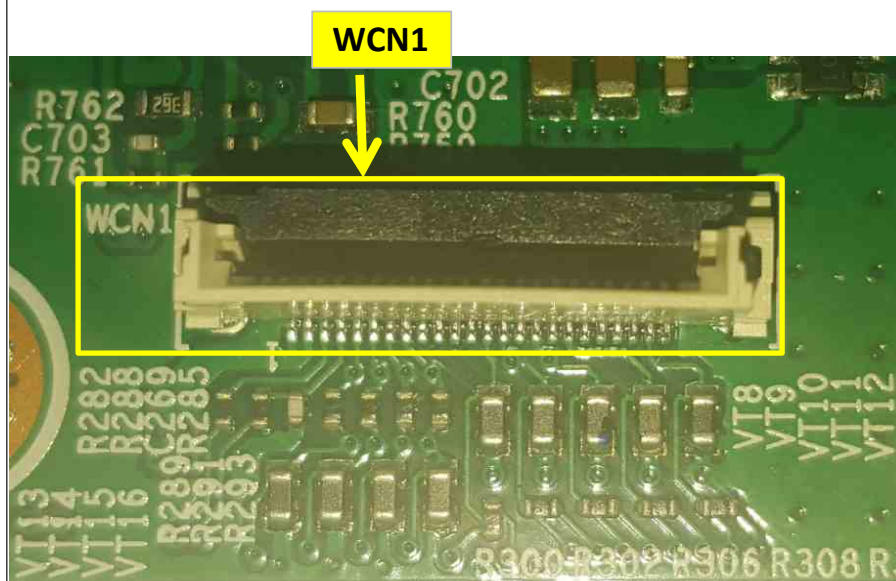


■ Location of Parts

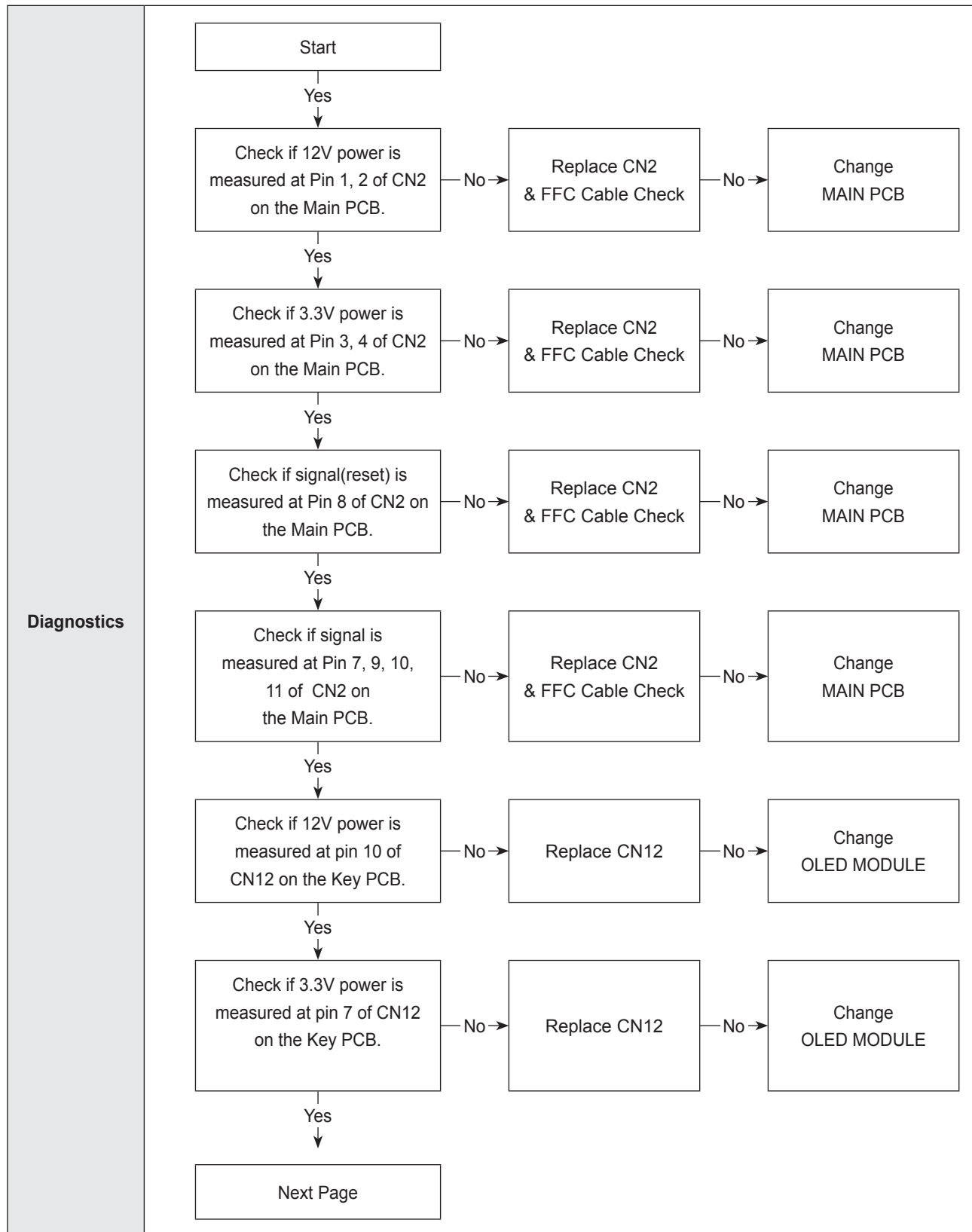
WIRELESS (5GHz Wireless Audio Transceiver)



WCN1	
MAIN↔Wireless	
1	W_+5V
2	W_+5V
3	DVCC33
4	DVCC33
5	W_IRQ
6	W_DARR_RST
7	W_SCL
8	W_SDA
9	NC
10	NC
11	NC
12	DGND
13	BCK
14	LRCK
15	NC
16	W_REAR_DATA
17	NC
18	W_C_WF_DATA
19	NC
20	NC
21	NC
22	NC
23	W_WP
24	DGND

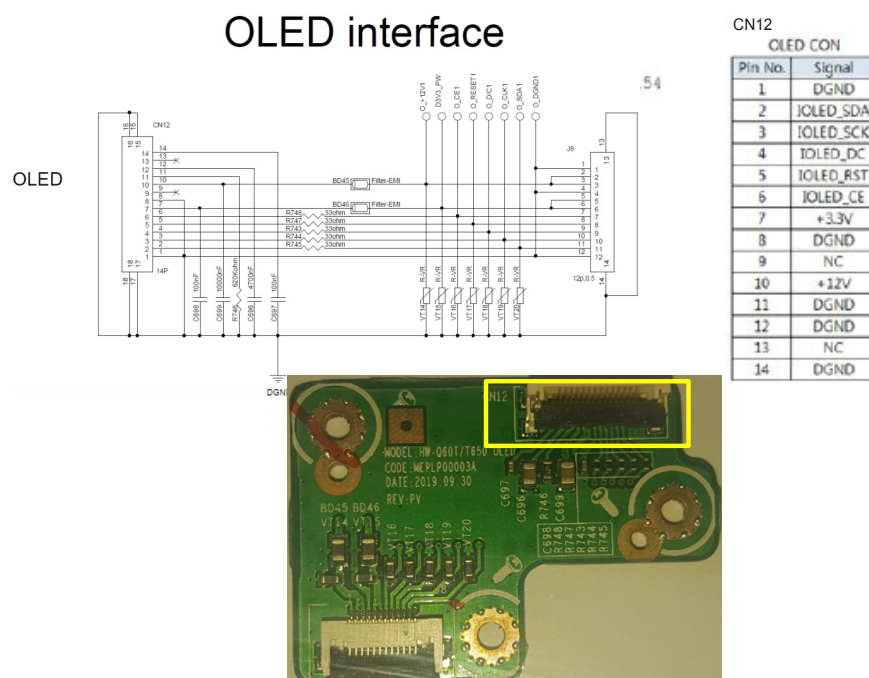
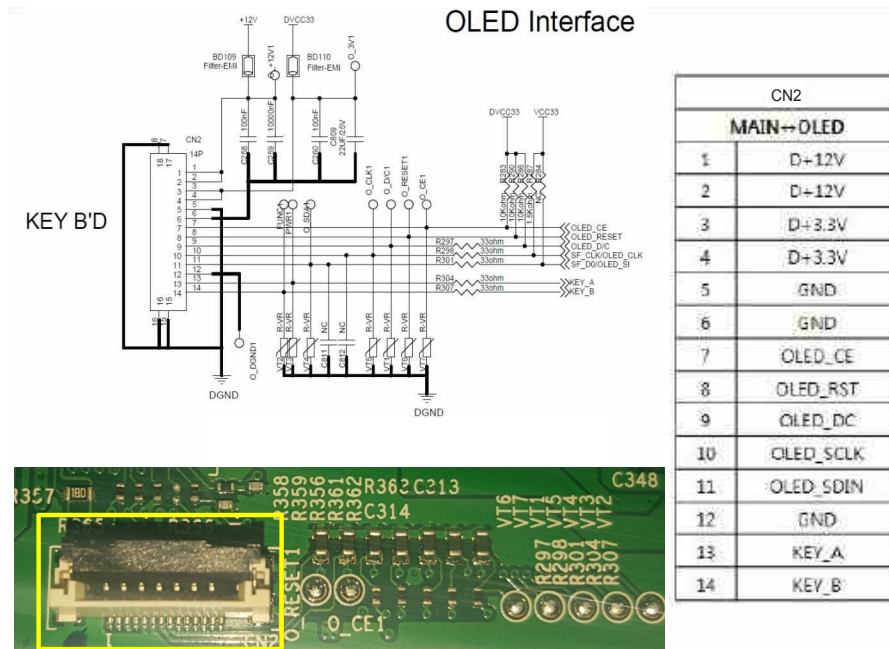


4-1-5. OLED Module Failure Analysis



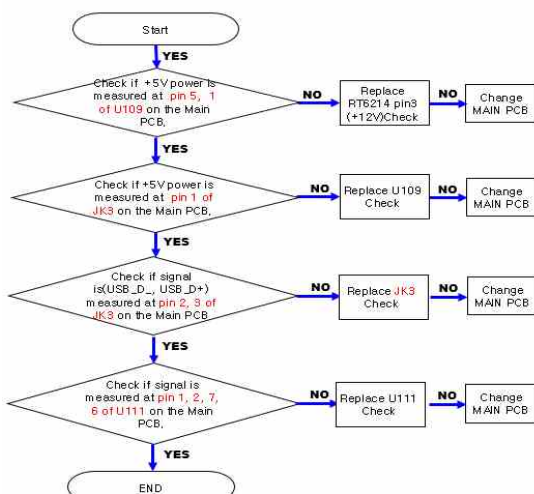
Diagnostics	Check if signal(reset) is measured at pin No.5 of CN12 on the Key PCB. No→Replace CN12No→Change OLED MODULE Yes↓ Check if signal is measured at pin No. 2, 3, 4, 6 of CN12 on the Key PCB No→Replace CN12No→Change OLED MODULE Yes↓ END
Caution	• Before working, disconnect the power cord.

■ Location of Parts

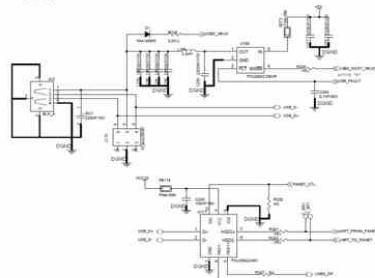


4-1-6. USB / HDMI Failure Analysis

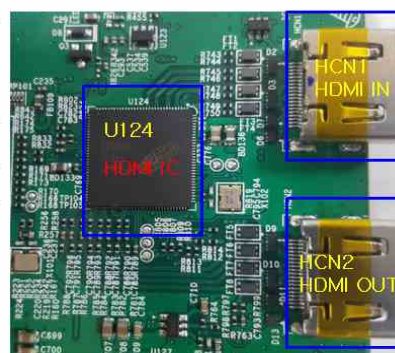
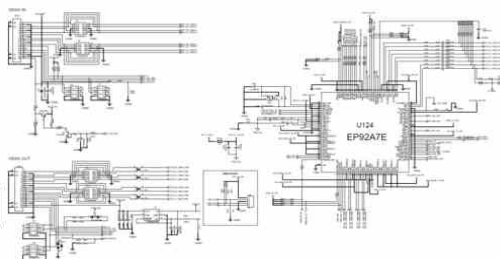
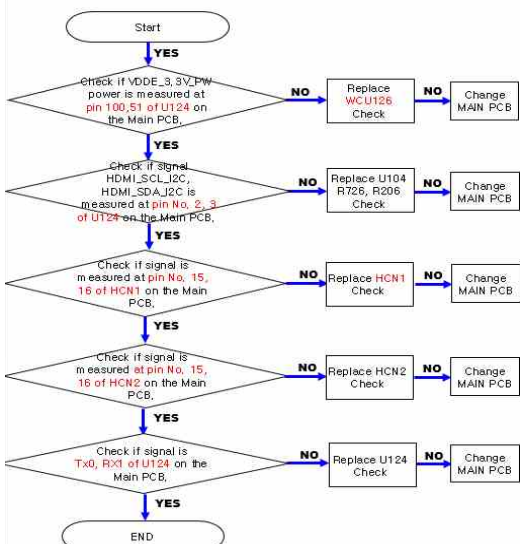
USB Failure Analysis



USB

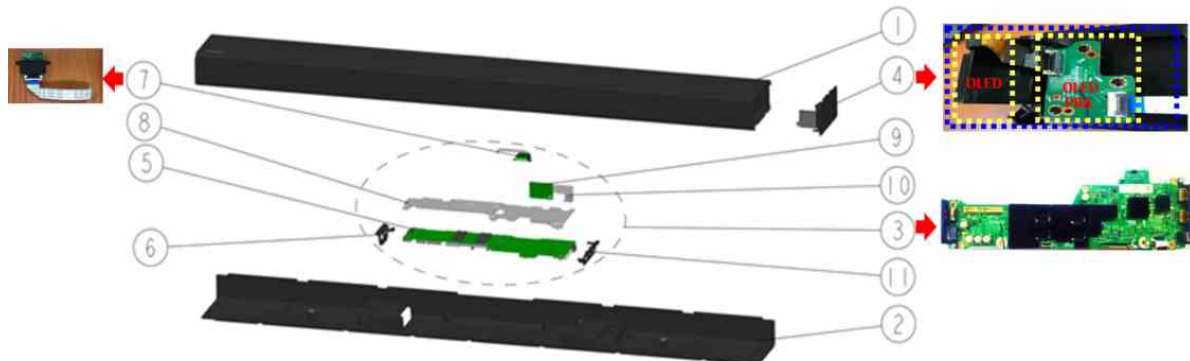


HDMI Failure Analysis



4-1-7. Exploded View & Parts (SET)

Exploded View & Parts (SET)



NO	PART CODE	Parts Name	Description & Specification
1	AH81-12713A	ASSY COVER TOP,DECO-BUTTON	ASSY COVER TOP,DECO-BUTTON
2	AH81-11737A	ASSY-COVER REAR	ASSY,HW-Q60T/ZA
3	AH81-11736A	ASSY MAIN PCB KIT	ASSY KIT,HW-Q60T/ZA
4	AH81-11751A	ASSY-COVER-SIDE,R	ASSY,HW-Q60T/ZA (including OLED Module + OLED PBA)
5	AH81-11739A	ASSY-PCB-MAIN	MAIN PCB ASSY
6	AH81-11740A	ASSY-COVER-USB	PC+ABS,V-0,NH-1017D / K21441
7	HEPOP00002A	ASSY PBA Bluetooth Module	ASSY PBA Bluetooth Module,Q60T
8	AH81-11743A	BRKT-MAIN	EGI-SECC,T0.8,NATURAL
9	AH81-09787A	ASSY-WIRELESS-MODULE	WSM620V,40.4x26.0x3.2mm
10	AH81-11744A	FFC-CABLE	BTYPE,24p,0.5mm Pitch,L80
11	AH81-11745A	ASSY-COVER-TERMINAL	PC+ABS,V-0,NH-1017D / K21441

4-2. Measures to be taken when the Protection Circuit operates

4-2-1. AMP Pre-Inspection relating to Power Protection

If you think there are problems at the AMP PCB, you can check the PCB without disassembling the set by following the test below.



Caution

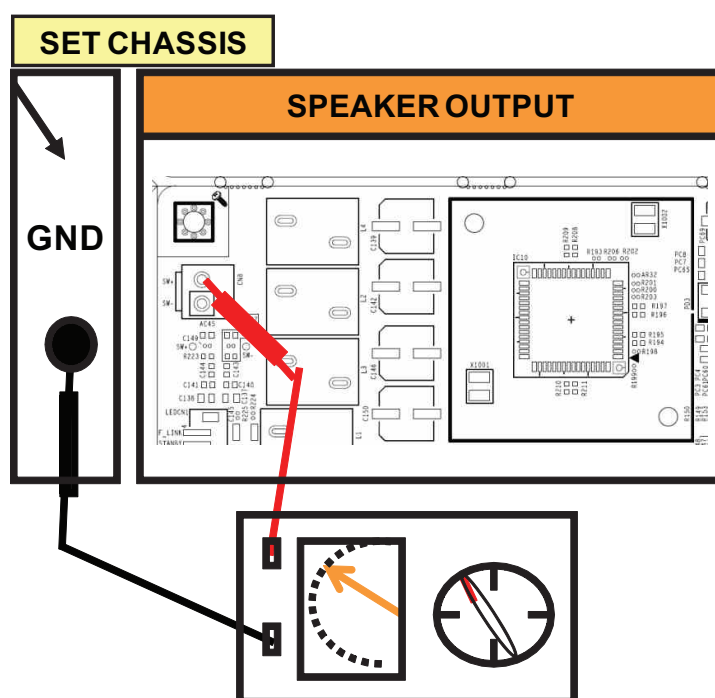
- Do not connect the power cord during the test!

Resistance using Tester

F/R CH

137K ohm

- If there is a large difference than the value listed above then the AMP PCB has a problem.



4-3. Initialization & Update

4-3-1. Set Initialization method

■ Sound Bar(Using the remote control)

1. Prepare the remote control.
2. Turn the unit off.
3. Press and hold the "Play/Pause" button on the remote for more than 5 seconds. 
4. If it was successful, "INIT"-">"OK" will be displayed for 2 seconds and be turned off.

■ Sound Bar(Without Using the Remote control)

1. Turn the unit on.
2. Press and hold the "+" button and "-" button on Sound Bar simultaneously for 5 seconds.
3. If it was successful, "INIT"-">"OK" will be displayed for 2 seconds and be turned off.
4. The Sound bar will be turned off



Top Panel

4-3-2. Pairing method

■ Sound Bar(Using the remote control)

1. Press and hold ID SET on the rear of the subwoofer(or Optional surround speakers) for least 5 seconds.
2. Prepare the remote control.
3. Turn the unit off.
4. Press and hold the "MUTE" button on the remote for more than 5 seconds. 
5. If it was successful, "ID SET" will be displayed for some seconds and be turned off.
6. When the soundbar is turned on, it will start finding wireless Subwoofer(or surround speakers)

■ Wireless subwoofer and rear speakers

1. Press ID Set button on the sub and rears.

※ Guide the customers to make sure each of components are in ID Set mode.

4-3-3. How to check the Firmware version

■ Software check method ... Sound Bar (Remote control use)

1. Prepare the remote control.
2. Turn the unit off.
3. Press the "SOUND CONTROL" button for 5 seconds. .
4. OLED will display each versions like below sequence. (DSP → HDMI → Tx → Rx)

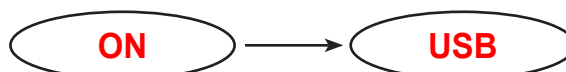
Tx : Wireless Tx module in the soundbar
Rx : Wireless Rx module in the rear kit



R2 VERSION DISPLAY
- NO CONNECT REAR : R2 ---
- CONNECT REAR : R2 XXX

4-3-4. USB Update procedure

- Step 1 : Power on the device. You can see the following OLED display.
Please Change Function "USB"



- Step 2 : Prepare the USB disk with update files then plug-in to the USB port.
You can see the following OLED display.

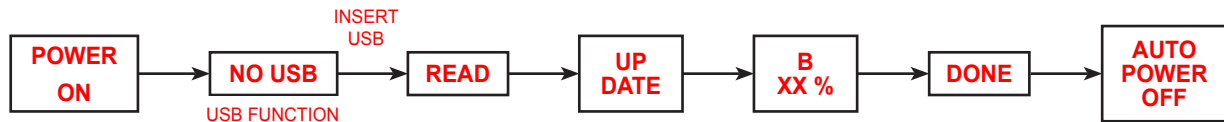


- Step 3 : Progressing update.

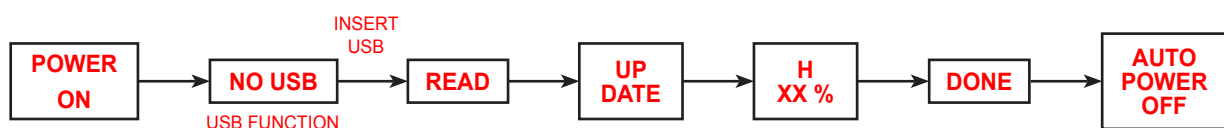
in case of Backend, HDMI and Wireless update, you can see the following OLED display.

4-3-5. USB Update procedure

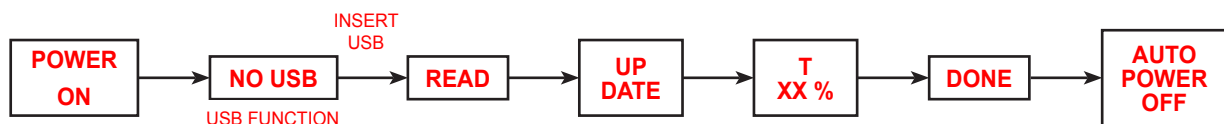
- Case 1 : Backend Update. (HMS-Q60TWWB-XXXX.X.bin)



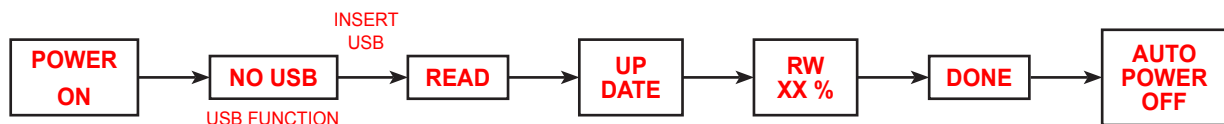
- Case 2 : HDMI Update. (HMS-Q60TWWH-XXXX.X.ssp).



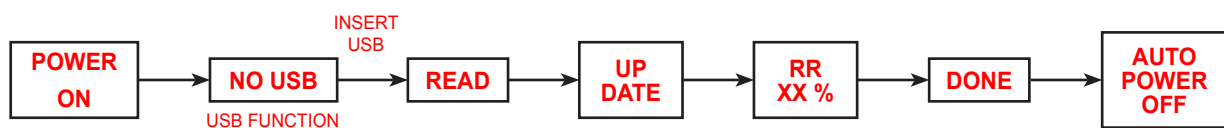
- Case 3 : Wireless Tx Module Update. (HMS-Q60TWWMR-T0XXX.ssp)



- Case 4 : Wireless SubWoofer Update. (HMS-Q60TWWMR-R0XXX.ssp)



- Case 5 : Wireless Rear Speaker Update. (HMS-Q60TWWMR-R2XXX.ssp)



ashakaas